Retrieval-Augmented Generation Engineering Notes

This document is a plain-text PDF used to exercise the PDF loading path. It contains no scanned images and no complex layout, so pure text extraction is sufficient.

Section 1. Why chunking matters

Retrieval quality is dominated by chunk granularity. Chunks that are too large mix several topics into one vector and dilute the semantic signal. Chunks that are too small fragment the answer and leave the language model without enough context to compose a coherent reply.

Section 2. Chunk size guidance

A practical default is three hundred to five hundred tokens per chunk, with an overlap of ten to twenty percent of the chunk size. The overlap exists so that a sentence landing exactly on a cut boundary still appears whole in one of the adjacent chunks.

Section 3. Structure aware chunking

Documents that carry explicit structure, such as Markdown headings or PDF chapter titles, give you free semantic boundaries. Prefer splitting on those boundaries over splitting on arbitrary character counts, because the author of the document already told you where the topics change.

Section 4. Metadata and citations

Every chunk should record its source file, its heading path, and its ordinal position. The heading path is the single most valuable piece of metadata: two sentences with identical wording mean different things in different chapters. Prefixing the heading path onto the chunk text measurably improves recall.

Section 5. Evaluation pitfalls

Evaluating a retrieval pipeline requires held out questions whose answers are not literally present in the corpus. If the answer appears verbatim in the document, recall metrics become meaningless because the retriever can match on surface form alone rather than on meaning.